

Has there been any side by side of how much water an ai prompt uses Vs. how much water it takes to make a pencil and paper?

Posted by u/Ok_Passion_6771 • 0 points • 9 comments

I'm trying to research this right now but I'm getting tons of weird obviously biased sources. I see a lot of the "ai uses so much water" and "pick up a pencil and paper!". And I've seen studies that say 40 prompts you put it use up the amount of water that comes out in one second of a shower. And then I saw a person say that and get downvoted to hell with replies like "but it's bad if a million people are doing it!"... They also love to encourage people to pick up a pencil and paper. Soooo, I'm just wondering how much water it takes to make one pencil and one paper (even though you can't usually buy them in that quantity so let's say 1 notebook and 1 pack of 3 pencils? I'm researching but maybe others are better than I am. And please cite sources. Thanks!

Comments

u/MelodicWallaby4476 • 2 points • 2025-08-15 17:30 UTC

Here are some links I have shared around before that seem relevant to your question:

<https://artinpraxis.org/the-environmental-impact-of-art-materials-sustainable-choices-for-conscious-creators/>

[https://en.wikipedia.org/wiki/Environmental_impact_of_paper]
(https://en.wikipedia.org/wiki/Environmental_impact_of_paper)

<https://calcedar.com/pencils-environmental-profile/>

[<https://momaa.org/environmental-impact-calculator-for-art-materials/>]
(<https://momaa.org/environmental-impact-calculator-for-art-materials/>)

From my experience however, I mostly do digital art nowadays, a piece that takes me 4-5 hours can be done on a local image generator at a higher quality in under a minute. The difference in total energy use from the computer alone is massive even without factoring in other conditions I would have used during the production (eg. overhead lighting, drinking water, table fan to keep cool).

On the grounds of environmental impact, genAI is generally better than most other forms of art with the

exception of things like human performance art or digital photography.

While I do art because I enjoy making art, the whole environmental argument is blatantly misconceptive and lacking in self-realization of one's own waste both immediate and factorial. I've said it many times, and I say this as an artist, but art is inherently wasteful AI or not. When comparing the expenditure of a new system you must directly weigh it against the costs of what it is competing with and in this discussion.

u/Ok_Passion_6771 • 2 points • 2025-08-16 03:45 UTC

Thank you for such a well thought out and well sourced answer. Things like this are what I was looking for with this question. Not to say that I'm just trying to confirm my own bias. I want to leave that out of it. I wanted to see things like you provided to sift through. I appreciate that you included sources which I'm now reading through. Your answer is respectful, resourceful and thought provoking. Thank you

u/bbt104 • 8 points • 2025-08-14 22:30 UTC

Thats the fun part. The pencil is almost impossible to calculate. Here's just a few factors that would have to be calculated together:

Growing of the tree that provided the wood

The water intake of the people who cut down the tree

The water intake of the mechanics who keep the equipment used to cut the tree down

The water used in the mining of the metals for the equipment

The water intake of the workers who mined the metals

The water intake of their mechanics

Repeat all of that but for the mining/making of graphite.

Thats all before the raw materials even reach the factory to be turned into a pencil.

Then theres the making of the boxes for said pencil

If the box has any plastic on it (think of the pencils sold in the same package style as batteries)

Then theres the transport of the pencil, if it came over seas thats even more calculations

Then theres the hidden cost of transporting fuel to truck stops so that the trucks hauling the pencils can refuel.

Then there the guys driving the truck.

Then the guys who load and unload the truck

Power for the lighting in the store

Then theres also the electricity used at the register

Then theres the fuel likely used by the person buying the pencil to get to the store, then return home.

There's a thousand more things that would need to be calculated, these are just the things I could think of off the top of my head.

There's a great quote by Milton Friedman:

>Look at this lead pencil. There's not a single person in the world who could make this pencil.

Remarkable statement? Not at all. The wood from which it is made, for all I know, comes from a tree that was cut down in the state of Washington. To cut down that tree, it took a saw. To make the saw, it

took steel. To make steel, it took iron ore. This black center—we call it lead but it's really graphite, compressed graphite—I'm not sure where it comes from, but I think it comes from some mines in South America. This red top up here, this eraser, a bit of rubber, probably comes from Malaya, where the rubber tree isn't even native! It was imported from South America by some businessmen with the help of the British government. This brass ferrule? [Self-effacing laughter.] I haven't the slightest idea where it came from. Or the yellow paint! Or the paint that made the black lines. Or the glue that holds it together. Literally thousands of people co-operated to make this pencil. People who don't speak the same language, who practice different religions, who might hate one another if they ever met! When you go down to the store and buy this pencil, you are in effect trading a few minutes of your time for a few seconds of the time of all those thousands of people. What brought them together and induced them to cooperate to make this pencil? There was no commissar sending ... out orders from some central office.

u/-paperbrain- • 1 points • 2025-08-16 11:22 UTC

If this is the kind of metric you're using though, even if you could find out the water consumption of each person and material process, you wouldn't compare that to ONLY the water used in cooling for AI, which are the common numbers that come up. To be parallel if that's your approach, you'd be looking for the water consumption in the construction of building the data centers, and the software company campuses where all of the engineers creating AI work, and all of their personal water consumption while doing it, and all of the water used to extract the materials to make all the computing and construction equipment and on and on.

The development of an AI system almost certainly involves the purchase of some pencils too and any water used in making and selling them

Now I know your point is sort of that it's incalculable if you go deep enough, and you're not wrong. But then that would lead one to throw up their hands and say no one can ever make an environmental claim. Who can say anything about carbon or water?

This is not just a problem of environmental impact. You could trace economic impact down an endless chain if you wanted. This way lies paralysis in doing anything about anything.

u/Ok_Passion_6771 • 1 points • 2025-08-16 23:39 UTC

So watcha wanna do? I turn my shower off early if I'm feeling like making a prompt.

u/[deleted] • 1 points • 2025-08-17 13:16 UTC

[removed]

u/Ok_Passion_6771 • 1 points • 2025-08-23 18:33 UTC

Okay we can start with the main things that use the most water. You first.

u/Ok_Passion_6771 • 3 points • 2025-08-14 23:36 UTC

Yes, exactly! It should be the same impossibility you speak of to calculate the water usage of someone's ai prompt. Yet, we can (maybe, depending on the source). Like both we can get an idea of it, on a large enough scale and long enough timeline. What I'm asking is, are the people who are against it because of "the environment" (usually this is the water argument) not the same ones telling ai users to use up MORE water through practicing by buying pencils and paper and drawing and throwing away and drawing and throwing away and practicing, etc. Someone with an idea for an image could use up 3 seconds of a shower, on a weekend, to communicate an idea they had. Not to mention the water or data centers processing their COMMENTS on these

platforms.

I just don't get it. I mean, I do... if you just think about the people against it being angsty. But all they need to do is look at how painting did not die because of photography. There will and has always been skilled people that draw and paint, since the first ever art was discovered. If they're ACTUALLY mad about how it will be harder for them to make money now... well that shit isn't news. Every art teacher I had said "90% of you won't make it." And that was in an art school you had to be good to even get into.

u/Interesting-Fox4064 • 3 points • 2025-08-14 22:16 UTC

Considering the water used by AI is used multiple times it's not really a close comparison

u/Ok_Passion_6771 • 5 points • 2025-08-14 22:49 UTC

I've gone through an incalculable amount of pencils and paper in my art career and I know at least 40-50 PERSONALLY throughout my life who have done the same. And we can include WATERcolor paintings and ALLLLLLL of that water being used to clean brushes and add to paint and the paper they use to create a thousand things that they will throw away in pursuit of that ONE thing they're looking for. All of these strangers being accused of taking water resources for a little thing they might like to do in their free time - a couple times a year or a month... are being told they are lazy for not using MORE water by practicing with pencil and paper, or paint and the water it takes to wash brushes or clean them... or am I not getting that right? Someone who doesn't know how to draw, doesn't have the time, but has an idea, that they can make in the FEW hours they have (between their job and family life and sleep and recreation, etc.) is harming things wayyyyy worse than millions of practicing artists who are going through pencils and paper and paint like a chain smoker?

u/nomic42 • 6 points • 2025-08-14 22:14 UTC

|Art Method|Estimated Water Use per Piece|Main Water Source (Materials vs. Energy)|Notes/Assumptions|

|:-|:-|:-|:-|

|Pencil on Paper|\~2–5 liters|Materials (paper production)|Based on recycled paper water footprint; assumes one A4 sheet and standard graphite pencil.|

|Acrylic Painting on Canvas|\~50–100 liters|Materials (paint & canvas production)|Acrylic paint manufacturing and cotton canvas water intensity.|

|Digital Art on Tablet|\~0.1–0.5 liters|Energy (electricity, device charging)|Based on renewable-heavy grid; excludes device manufacturing.|

|Digital Art via AI (Data Center)|\~5–15 liters|Energy (data center cooling)|Based on average AI model inference water footprint per image.|

|Oil Painting on Linen|\~150–300 liters|Materials (linen, oil-based paints)|Linen production is highly water-intensive; includes paint manufacturing.|

|Clay Sculpture (Fired)|\~20–50 liters|Materials (clay processing) & Energy (kiln firing)|Includes mining, processing, and firing energy costs.|

|Charcoal Drawing|\~2–4 liters|Materials (paper production)|Assumes one sheet A4 and charcoal stick manufacturing.Art

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u/FionaSherleen • 2 points • 2025-08-16 04:16 UTC

Where is the source it used? 15 liters is bull.

u/Ok_Passion_6771 • 1 points • 2025-08-23 18:34 UTC

The thread is open to your rebuttal

u/Ok_Passion_6771 • 3 points • 2025-08-14 22:55 UTC

I don't trust ai right off the bat. For every 100 things it gets right, There is still (probably) a discrepancy or a spiders Gorge outlier it's counting for. But it's the same as trusting Wikipedia. And Wikipedia only got better. I think it'd be cool if ai would cite what it referenced when it gave you something, the same way you can view sources in Wikipedia.